

Total No. of Questions : 8]

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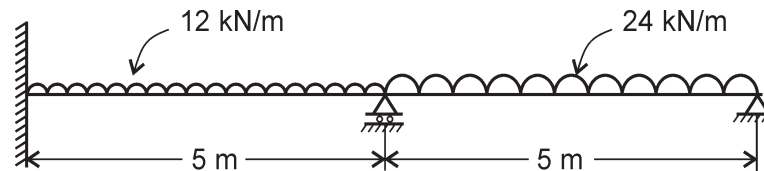
Roll No

MI-7102 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Numerical Method in Mining Engineering
Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.

1. What is the applications of finite elements methods in geomechanics?
2. Explain the followings:
 - i) Simulation of joints in strata
 - ii) Linear and non linear analysis in FEM
3. Give short notes on:
 - i) Homogeneous
 - ii) Isotropic
4. Analysis the Beam shown by FEM and determine the end reactions.



5. Give step by step finite difference method of solving mining related problems take a suitable hypothetical problem and discuss method of solving it by FDM.
6. Write a short note on finite element method with advantage and disadvantages of FEM.
7. Describe the method of numerical modelling of discontinuous rock mass.
OR
Give a detailed comment on the application of finite element method in mining engineering.
8. Write short notes on any two of the following.
 - a) Non-linear elastic and elasto plastic.
 - b) Discretisation.
 - c) Slope stability.
 - d) Element stiffness.
